

Classifying Kidney Transplant Rejection from Biopsy RNA

Transplant rejection and kidney biopsy

- **Rejection:** The recipient's immune system attacks the donated kidney.
- **Kidney biopsy:** A small tissue sample collected from the kidney for examination.
 - **Histology:** Microscopic examination reveals injury and inflammation and supports the recorded diagnosis.
 - **Molecular measurements:** Counts of selected RNA types provide the model's inputs.

Possible Use: Molecular Second Opinion for Ambiguous Biopsies

- Sometimes the microscopic findings are borderline or conflict with other clinical evidence.

Hypothetical Example:

A biopsy shows mild inflammation, but the microscopic findings do not clearly establish rejection.

- I used RNA to classify recorded diagnoses in this personal project.

Hypothetical Example, continued:

A high rejection score could add evidence for specialist review of this biopsy.

Evidence for a molecular second opinion

Banff guidance 2026

Considers validated molecular tests for difficult biopsy interpretations

B-HOT study Rosales, 2022

Higher initial molecular scores in patients who later developed chronic active antibody-mediated rejection, despite no initial diagnosis of it

Why incorrect rejection calls matter

Unnecessary rejection treatment can weaken the body's defenses and worsen an infection.

KDIGO (2009); BK polyomavirus consensus (2024)

This personal project measured agreement with recorded diagnoses. Its value in ambiguous biopsies needs direct evaluation.

What the dataset contains

1,395 biopsy specimens, with one row per specimen in each table.

RNA counts

Column	Example
Specimen ID (key)	GSM6510425
IFNG	4
GUSB (reference)	260
Other RNA counts	...

1 : 1

Same ID

Specimen information

Column	Example
Specimen ID (key)	GSM6510425
Recorded diagnosis	No rejection
Study group	Discovery
Other details	...

- Includes 1,193 transplant biopsies and 202 native-kidney controls.
- The study does not document separation by patient or referring center between study groups.

Outcome: rejection versus no rejection

The model combines the three rejection categories in the public dataset.

Recorded diagnosis	Meaning	Model label
Antibody-mediated rejection	Antibodies: immune proteins	Rejection
T-cell-mediated rejection	T cells: immune cells	Rejection
Mixed rejection	Both rejection processes	Rejection
No rejection	No recorded rejection	No rejection

The authors excluded some diagnostic categories from the dataset.

The measurements used to build the features

770 RNA counts per specimen, one for each RNA type measured.

758

Model measurements

- Include human and viral RNA.
- Signals depend on gene activity and the cell mixture.

12

Housekeeping references

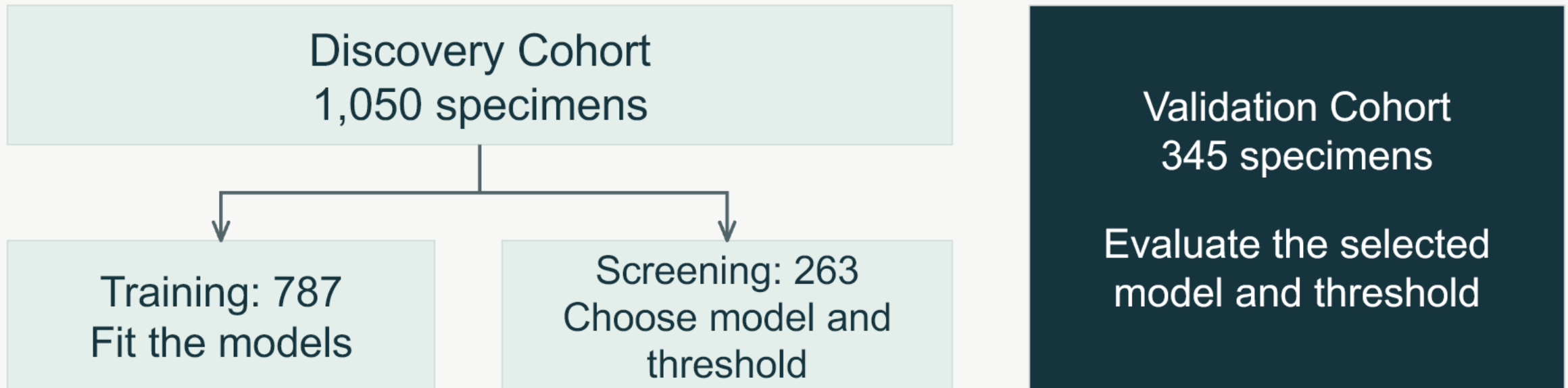
- 12 different reference RNAs.
- Used to adjust for differences in how much RNA each specimen supplies.

Preparing counts for the model

For each of the 758 RNA counts	IFNG example
1. Start with the raw count	4
2. Calculate $\log_2(\text{count} + 1)$	2.322
3. Average $\log_2(\text{count} + 1)$ for the 12 reference RNAs	7.900
4. Subtract that reference	$2.322 - 7.900 = -5.578$

Development and evaluation specimens

- I used the authors' two study groups: discovery and validation cohorts.
- My training / screening split kept diagnosis proportions similar.



Models compared

Each model produces a score for rejection versus no rejection.

Model	What it learns
Constant baseline	One score: the training rejection fraction
IFNG only (Logistic regression)	One immune-related RNA measurement as a simple benchmark
All RNA (Logistic Regression)	A weighted combination of 758 measurements. Regularization limits the size of the weights.
CatBoost	A combination of decision trees

Choosing a rejection threshold

Recall

Recorded rejection cases detected

All recorded rejection cases

(False negatives are missed cases.)

Precision

Flags with recorded rejection

All positive flags

(False positives are incorrect flags.)

Selection using screening specimens

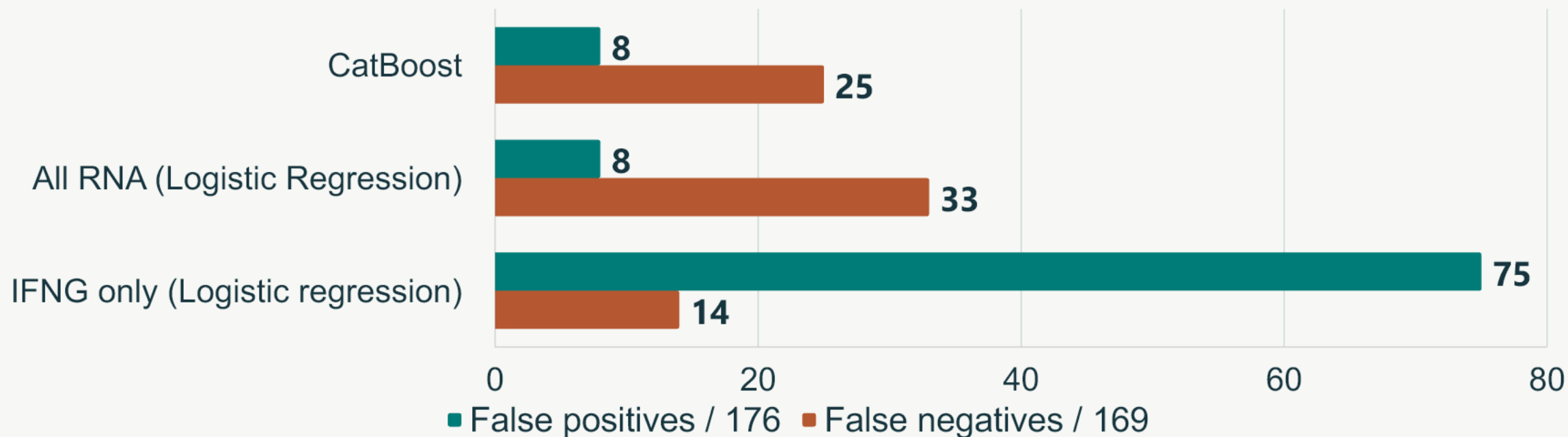
Step	Rule
1	Detect at least 90% of recorded rejection cases.*
2	Among qualifying choices, minimize false positives.
3	Keep the chosen model and threshold for evaluation.

*90% is an experimental target.

Source: discovery screening rule and selected model metadata

Validation results

Same 345 specimens: 169 rejection and 176 no rejection



- Constant baseline: 0 false negatives, 176 false positives. It flags everyone.
- CatBoost versus IFNG: 67 fewer false positives, 11 more false negatives.

Most CatBoost misses were T-cell rejection

144 / 169 detected

85.2% recall (sensitivity)
in evaluation

- Screening: 157 / 174 detected (90.2%).
- Evaluation falls below the 90% selection target.

Misses by recorded diagnosis

Recorded rejection	Missed / total
Antibody-mediated	6 / 56
T-cell-mediated	18 / 95
Mixed	1 / 18

The advantage over logistic regression is uncertain

CatBoost versus All RNA (Logistic Regression)

Observed recall difference: +4.73 percentage points (8 / 169 specimens).

How I calculated the 95% confidence interval

1. I drew 2,000 resamples of the 345 validation specimens, with replacement.
2. I kept both models and thresholds fixed and compared recall on each draw.
3. The middle 95% ran from -0.01 to $+9.74$ percentage points, including zero.

Why I also repeated model development

One training / screening split might favor CatBoost.

I checked whether changing those assignments changed the selected model.

Model choice depended on the development split

I repeated development with 20 splits of the 1,050 discovery specimens.

Each random split kept the four diagnosis proportions similar.

I used the same rows for both model families.

630 fitting
Train both model families

210 screening
Choose model and threshold

210 assessment
Measure held-out errors

Screening: at least 90% recall, then fewest false flags.

I used ROC-AUC to break ties.

CatBoost	All RNA (Logistic Regression)
13 / 20 selections	7 / 20 selections

The splits shared specimens; the 20 selections were not independent trials.

Prediction service and demonstration

- CSV input through the command line or web application.
- Training and prediction share the preparation code.



- A valid request returns the specimen ID, model score, threshold, flag and model version.
- Missing or duplicate measurements, negative counts or non-finite values stop scoring.

One public specimen through the service

- GSM6510425: recorded no rejection.

Model score	Selected threshold	Rejection flag
0.274934	0.876588	False (below cutoff)

Model: 20260915_shared:any_rejection:catboost_all_depth4

- The complete input produces a flag that agrees with the recorded diagnosis.
- Removing IFNG clears the previous result and returns no new score.

“Missing required assay targets: IFNG”

The service reproduces all 345 saved scores

- API, command line and saved evaluation agree within 10^{-12} .

Check	Observed behavior
Reordered columns	Same model score
Invalid or incomplete counts	No score
Incompatible model metadata	Model does not load

- 94 tests passed, including these failure cases.
- The local container was checked. Cloud deployment is deferred.

Possible next steps

Ideas for extending this personal project

Question	Possible approach
Would the results hold in another dataset?	Test the saved models and thresholds on new transplant biopsies. Record patient, center and lab quality information.
Would RNA help with uncertain biopsies?	With clinical collaborators, compare assessment with and without the RNA score. Review decisions and follow-up outcomes.

What this project accomplished

Model comparison

CatBoost, 345 evaluation specimens

14.8% missed

25 of 169 recorded rejection specimens

4.5% false flags

8 of 176 recorded no-rejection specimens

Fewer false flags than IFNG, but more misses.

Advantage over logistic regression remains uncertain.

Prediction service and demo

- Shared preparation code reproduced all 345 saved scores.
- Working demo with valid input and clear error messages.

Backup: measuring biopsy RNA with NanoString nCounter

- RNA includes messages cells make when genes are active.
- An assay is a laboratory test. B-HOT is the fixed panel of RNA types measured in the public dataset.
- The laboratory extracts RNA from preserved tissue. Probes recognize selected RNAs, and NanoString nCounter counts their identifying barcodes.

Example: IFNG count = 4 in one public specimen

Backup: complete primary comparison

Model	Misses	False flags	Sensitivity	Specificity	ROC-AUC
CatBoost	25	8	85.2%	95.5%	0.969
All RNA (Logistic Regression)	33	8	80.5%	95.5%	0.956
IFNG only (Logistic regression)	14	75	91.7%	57.4%	0.842
Constant	0	176	100.0%	0.0%	0.500

Denominators: 169 rejection; 176 no rejection. Fixed thresholds per model.

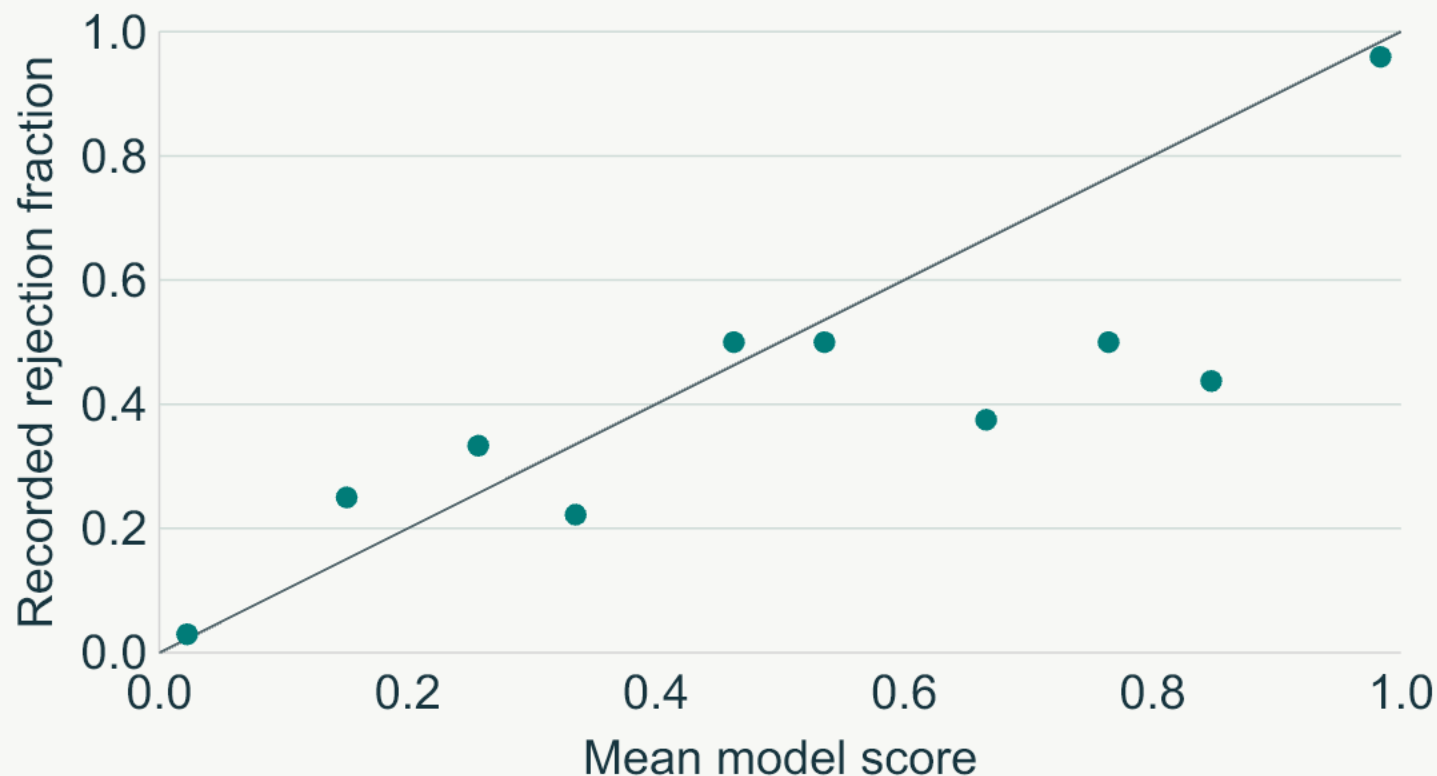
Backup: normalization and model settings

Normalized measurement = $\log_2(\text{raw count} + 1)$
– mean of the 12 housekeeping $\log_2(\text{count} + 1)$ values

Model	Additional preparation and settings
All RNA (Logistic Regression)	Subtract each feature's training mean, then divide by its training standard deviation. Reuse these values for later specimens. Regularization: $C = 0.1$.
CatBoost	No extra feature scaling. 300 trees, depth 4, learning rate 0.04.

Backup: interpreting model scores

CatBoost on 345 validation specimens



Score bin 0.8–0.9

- Mean score: 0.847
- Recorded rejection: 7 / 16
- Observed fraction: 43.8%
95% interval: 23.1–66.8%

Diagonal: score equals observed rejection fraction

Backup: rejection subtype follow-up

Model	Any-rejection misses	False flags
Binary CatBoost	25 / 169	8 / 176
Four-class CatBoost	29 / 169	8 / 176
Four-class logistic	28 / 169	10 / 176
Separate component models	24 / 169	15 / 176

- Four-class logistic recognized 10 / 18 mixed diagnoses, with 5 false mixed calls among 327 non-mixed specimens.

Backup: dataset, assay and deployment limits

Question	Evidence and remaining limits
Patient / center separation?	Not documented. All specimens were processed at one laboratory.
Study population?	Includes 202 native-kidney controls. Selected ambiguous diagnoses were excluded.
Laboratory assay quality?	File checks were tested. Laboratory quality was not independently reassessed.
Individual probabilities?	Score calibration remains limited.
Cloud deployment?	Local container tested. Cloud work deferred.

Backup: sources and contributions

Public study

- Zhang et al., Laboratory Investigation (2024): GSE212160 raw NanoString B-HOT measurements.

This personal project

- Binary model comparisons and runs produced here, shared preparation code, and a tested local prediction service.

Presentation guidance

Bourne (2007), Kosslyn et al. (2012), Garner and Alley (2013), Rougier et al. (2014)

Backup: UNOS research connection

- UNOS researchers studied anatomical issues in donor kidney photographs.
- Labels came from records of transplantation or refusal because of anatomy concerns.
- The intended use is to support clinical assessment and improve consistency.
- The connection: use recorded assessments to develop additional evidence for specialists.

This is a different clinical task and does not validate my classifier.